

On the Asymptotic Behavior and Multivalued Nature of the n -th Roots of i

Pearl Bipin Pulickal

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Abstract

This paper provides a rigorous and pedagogically-focused analysis of the n -th roots of the imaginary unit, i . We explore the multivalued nature of complex roots, deriving the general formula for the n distinct roots using De Moivre's theorem. A deeper insight is offered through the lens of the complex logarithm, elucidating the concept of branch cuts and the principal value. Finally, we address the asymptotic behavior as $n \rightarrow \infty$, demonstrating that while the set of roots becomes dense on the unit circle, the sequence of principal roots converges unambiguously to 1.

1 Introduction

In the realm of real numbers, the operation of taking an n -th root of a positive number yields a single positive result. The complex plane, however, is a far richer environment. The fundamental theorem of algebra guarantees that any polynomial of degree n has exactly n complex roots. The expression $z = w^{1/n}$ is therefore equivalent to solving the equation $z^n - w = 0$, which must have n solutions for z .

The question "What is the n -th root of i ?" is thus ill-posed without further qualification. The answer is not a single number, but a set of n distinct complex numbers. This paper aims to:

1. Rigorously define the set of all n -th roots of i .
2. Provide insight into their geometric arrangement in the complex plane.
3. Use the complex logarithm to explain the origin of this multiplicity and define the principal root.
4. Analyze the convergence of the roots as n approaches infinity, resolving the initial query about their limiting behavior.

2 Polar and Exponential Representation of i

To perform root and power operations in the complex plane, it is vastly more convenient to move from the Cartesian form ($z = x + iy$) to the polar or exponential form ($z = re^{i\theta}$).

The imaginary unit i has Cartesian coordinates $(0, 1)$. Its magnitude r (distance from the origin) and argument θ (angle from the positive real axis) are:

$$r = |i| = \sqrt{0^2 + 1^2} = 1$$

$$\theta = \arg(i) = \frac{\pi}{2}$$

A critical feature of the complex argument is its periodic nature. Adding any integer multiple of 2π (a full rotation) to the angle does not change the position of the point in the plane. Therefore, the argument of i can be expressed more generally as:

$$\arg(i) = \frac{\pi}{2} + 2\pi k, \quad \text{for any integer } k \in \mathbb{Z}$$

This multiplicity of the argument is the fundamental source of the multiple values for roots.

Using Euler's formula, $e^{i\theta} = \cos(\theta) + i\sin(\theta)$, we can write the general exponential form of i as:

$$i = 1 \cdot \exp \left[i \left(\frac{\pi}{2} + 2\pi k \right) \right], \quad k \in \mathbb{Z} \quad (1)$$

3 Deriving the Set of n-th Roots

We seek to find all complex numbers z_k such that $z_k^n = i$. Using our exponential form, we can write:

$$z_k = i^{1/n} = \left(\exp \left[i \left(\frac{\pi}{2} + 2\pi k \right) \right] \right)^{1/n}$$

By the properties of exponents, which for complex numbers is encapsulated in De Moivre's theorem, we have:

$$z_k = \exp \left[\frac{i}{n} \left(\frac{\pi}{2} + 2\pi k \right) \right] = \exp \left[i \left(\frac{\pi + 4\pi k}{2n} \right) \right] \quad (2)$$

These are our roots. They all have a magnitude of $|z_k| = 1$, so they lie on the unit circle. To find the number of *distinct* roots, we can substitute consecutive integer values for k .

Let's test $k = 0, 1, 2, \dots, n-1$:

- $k = 0$: $z_0 = \exp \left[i \left(\frac{\pi}{2n} \right) \right]$
- $k = 1$: $z_1 = \exp \left[i \left(\frac{5\pi}{2n} \right) \right]$
- $k = 2$: $z_2 = \exp \left[i \left(\frac{9\pi}{2n} \right) \right]$
- ...
- $k = n-1$: $z_{n-1} = \exp \left[i \left(\frac{\pi + 4\pi(n-1)}{2n} \right) \right]$

These n roots are all distinct, as their arguments are all different and lie within a 2π interval. If we test $k = n$:

$$z_n = \exp \left[i \left(\frac{\pi + 4\pi n}{2n} \right) \right] = \exp \left[i \left(\frac{\pi}{2n} + 2\pi \right) \right] = \exp \left[i \left(\frac{\pi}{2n} \right) \right] = z_0$$

The cycle repeats. Therefore, there are exactly n distinct roots, generated by letting $k = 0, 1, 2, \dots, n-1$.

4 Logarithmic Insights and Branch Cuts

The multivalued nature of roots is elegantly formalized by the complex logarithm. The logarithm of a complex number $z = re^{i\theta}$ is defined as:

$$\log(z) = \ln(r) + i(\theta + 2\pi k), \quad k \in \mathbb{Z}$$

where $\ln(r)$ is the standard real natural logarithm.

To make the function single-valued, we define a **principal value** of the logarithm, denoted $\text{Log}(z)$, by restricting the argument to a specific interval, typically $(-\pi, \pi]$. This corresponds to setting $k = 0$ and choosing the unique angle in that range. This restriction necessitates a **branch cut**, typically along the negative real axis, which we agree not to cross to maintain continuity.

We can define complex exponentiation using the logarithm: $w^c = \exp(c \log w)$. Applying this to $i^{1/n}$:

$$i^{1/n} = \exp\left(\frac{1}{n} \log(i)\right)$$

Substituting the multivalued logarithm of i (where $r = 1, \theta = \pi/2$):

$$\log(i) = \ln(1) + i\left(\frac{\pi}{2} + 2\pi k\right) = i\left(\frac{\pi}{2} + 2\pi k\right)$$

Thus,

$$i^{1/n} = \exp\left(\frac{1}{n} \cdot i\left(\frac{\pi}{2} + 2\pi k\right)\right) = \exp\left[i\left(\frac{\pi + 4\pi k}{2n}\right)\right]$$

This reproduces our result from Section 3, clarifying that the set of roots arises from the multivalued nature of the complex logarithm.

The **principal n-th root** is defined by using the principal value of the logarithm (i.e., setting $k = 0$).

$$\text{Principal Root of } i^{1/n} = \exp\left(\frac{1}{n} \text{Log}(i)\right) = \exp\left(\frac{1}{n} \cdot i\frac{\pi}{2}\right) = \exp\left(i\frac{\pi}{2n}\right)$$

This corresponds to our root z_0 .

5 Convergence Analysis as $n \rightarrow \infty$

We can now rigorously analyze the behavior of the roots as n becomes arbitrarily large. It is crucial to distinguish between the convergence of the *sequence of principal roots* and the behavior of the *set of all roots*.

5.1 Convergence of the Principal Root

Let's consider the sequence of principal roots, $\{z_0^{(n)}\}_{n=1}^{\infty}$, where $z_0^{(n)}$ is the principal n -th root of i .

$$z_0^{(n)} = \exp\left(i\frac{\pi}{2n}\right) = \cos\left(\frac{\pi}{2n}\right) + i\sin\left(\frac{\pi}{2n}\right)$$

We take the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} z_0^{(n)} = \lim_{n \rightarrow \infty} \left[\cos\left(\frac{\pi}{2n}\right) + i\sin\left(\frac{\pi}{2n}\right) \right]$$

Since $\cos(x)$ and $\sin(x)$ are continuous functions, we can move the limit inside:

$$\begin{aligned} &= \cos\left(\lim_{n \rightarrow \infty} \frac{\pi}{2n}\right) + i \sin\left(\lim_{n \rightarrow \infty} \frac{\pi}{2n}\right) \\ &= \cos(0) + i \sin(0) = 1 + 0i = 1 \end{aligned}$$

Thus, the sequence of principal n -th roots of i converges to 1.

5.2 Behavior of the Set of All Roots

For any finite n , the set of roots $\{z_k\}_{k=0}^{n-1}$ consists of n points, evenly spaced on the unit circle with an angular separation of $\frac{2\pi}{n}$.

As $n \rightarrow \infty$, two things happen:

1. The number of roots, n , goes to infinity.
2. The angular separation between adjacent roots, $\frac{2\pi}{n}$, goes to zero.

The collection of all n -th roots for large n does not converge to a single point. Instead, the set of roots becomes **dense** on the unit circle. For any point on the unit circle, one can find a sequence of roots (by varying both k and n) that converges to it. The "limit" of the set of roots is the entire unit circle itself.

6 Conclusion

The expression $i^{1/n}$ is not a single value but a set of n distinct complex numbers lying on the unit circle. This multivaluedness is a direct consequence of the periodic nature of the argument in the complex plane, a concept formalized by the complex logarithm and the necessity of branch cuts.

While the collection of all n -th roots spreads out and becomes dense on the unit circle as $n \rightarrow \infty$, the clearly defined *principal root* follows a convergent path. The sequence of principal n -th roots of i converges unequivocally to the real number 1. This analysis resolves the initial skepticism by distinguishing between the behavior of a specific sequence and the limiting distribution of an entire set of values, showcasing the subtlety and rigor required in complex analysis.